

Launching a CubeSat: Rules, laws, and best practice

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Joost Vanreusel

European Space Agency, ESTEC, Strategy Department, Noordwijk, The Netherlands

1 CubeSats: A different class of satellites?

Considering their standardized form factor and dedicated technical features and interfaces, from a technical viewpoint, CubeSats form a specific class of satellites, governed by dedicated engineering standards, new or tailored test methodologies, standardized launch adapters, evolved approaches to product assurance, and contributed to a paradigm shift in the launcher sector. Compared with larger, traditional satellites, CubeSats typically have relatively short development cycles, smaller engineering teams, and lower production and operational costs. Additionally, CubeSats often involve players that are relatively new to space activities, including non-governmental actors such as academic institutions and private companies [1].

On the other hand, while some nations have adopted clauses in the national space legislation specifically applicable to pico-, nano-, or microsatellites, from the perspective of public international law, CubeSats are not treated as an individual class or type of satellites. They are bound to the same national and international laws applicable to conventional spacecraft with bigger masses. Bringing a satellite into use, independently from its type or size, is an official national space activity, and it is up to the entities that develop and launch small satellites, including CubeSats, to ensure compliance with the applicable laws. For example, rules and regulations for communication with space objects are legally binding in all member states of the International Telecommunication Union (ITU) [2]. Additionally, those states that have ratified the United Nations space treaties, which are explained in Section 2, must conduct space activities in accordance with applicable binding norms.

In support of a sustainable and responsible use of outer space, it should be recognized that legal and regulatory aspects form an integral part of satellite project management. These legal and regulatory aspects also apply to CubeSats developed by private or public entities. This may create an extra challenge as certain processes may appear to be difficult for less-experienced developers or even for government organizations that are not used to implement the related practice. Due to the typical characteristics of CubeSats, that can be relatively modest in complexity, size, and cost when compared with conventional satellites, they have been the first ever satellite from a given state to be launched into space, in multiple cases [3]. In particular, developers and governments from states new to the cluster of states undertaking space activities may not be fully aware of the applicable legal norms. Those include the

obligation to set up a national register of space objects or the obligation for the state to keep oversight over national space activities undertaken by private parties.

Several important initiatives in this area have been undertaken at an international level, among others, through the World radiocommunication conferences, in the frame of which experts have investigated the dedicated regulations for the assignment of radio frequencies to small satellites. The United Nations Committee on the Peaceful Uses of Outer Space (UNCOPUOS) addresses issues of privatization and commercialization of space activities, which resulted in UN resolutions on the concept of “*launching state*” [4] and in recommendations for states when enacting regulatory frameworks for national space activities [5,6]. For small satellite developers the Secretariat of the UN Office on Outer Space Affairs (UNOOSA) has developed, in consultation with ITU, an information handout on small satellites [7]. Appreciating the relevant legal terminology and being aware of the formal obligations coming forward from applicable rules and laws are key for CubeSat developers. These related processes form an integral part of the project planning and management, and their effective implementation may prevent issues during the final phases of the mission before launch and during the spacecraft operations.

This chapter is organized as follows: [Section 2](#) introduces the main space treaties; [Section 3](#) describes the issues of launch authorization; [Section 4](#) deals with the radio frequencies allocation; [Section 5](#) describes the steps of spacecraft registration; safety aspects of a CubeSat launch are treated in [Section 6](#); [Section 7](#) is dedicated to export issues; [Section 8](#) talks about space debris and collision avoidance; eventually, [Section 9](#) is concerned with third-party liability issues.

2 Space law

During the dawn of space exploration, while opposing powers were racing for new accomplishments in space to demonstrate their superiority, the need emerged to regulate the exploration and exploitation of outer space and its celestial bodies. During the 1960s, 1970s, and 1980s, several international treaties and agreements were adopted. They later formed the basis for amendments and clarifications, and the treaty obligations have been specified and implemented through national legal frameworks by several countries around the world. Analogous to the law of the sea that governs the rights and duties of states in maritime waters, a set of internationally legally binding treaties, norms, and principles serve to regulate spaceflight undertakings for scientific, commercial, or other purposes. The underlying principle of the *Outer Space Treaty* [8], which entered into force in 1967, is the prohibition of weapons of mass destruction in space and the limitation to utilize the Moon and other bodies for peaceful purposes only, without appropriation. It furthermore outlines States’ responsibilities in governing national space activities (see [Section 3](#)). The *Liability Convention* [9], which came into force in 1972, establishes liability rules for launchers and spacecraft on ground, during launch and in outer space (see [Section 9](#)), and the *Registration Convention* [10] from 1976 outlines a system to notify, identify, and register space missions (see [Section 5](#)). Other major space treaties are the *Rescue Agreement*, from 1968,

governing the rescue and return of astronauts by any state, and the *Moon Agreement* that entered into force in 1984, overseeing the exploitation of the Moon and other bodies and any stations placed there. While certain activities may be subject to specific requirements, the space treaties apply to any type of satellite as its definitions of space objects make no distinction in type, size, or function [5]. The set of the first three treaties mentioned earlier is crucial, together with any applicable national law, to comprehend the legal framework in which small satellite missions are assumed.

Space law: Practical recommendations for CubeSat developers

- Familiarize with the space treaties relevant to the mission, primarily the *Outer Space Treaty*, the *Registration Convention*, and the *Liability Convention*. The treaties are freely available online.
- Verify which of the treaties have been ratified by the State. A list of states party to the space treaties can be consulted on the website of the United Nations Office on Outer Space Affairs (UNOOSA).
- Check if the activity is subject to additional national (space) legislation or regulatory acts and familiarize with the requirements and procedures. A collection of laws relating to exploration and use of outer space is presented on the website of the United Nations Office for Outer Space Affairs.

3 Licensing and mission authorization

Placing a satellite in orbit requires a number of subsequent and formal approvals at various levels from different actors, which vary per country. On the side of the developer and customer, the mission is generally submitted to subsequent technical reviews at different stages of development, typically concluding with a flight acceptance review or equivalent, to declare the satellite and its ground segment ready for launch and operations. The launch authority, being the entity overseeing the launcher and offering the launch service to deploy the spacecraft to orbit, has to clear the spacecraft and the launcher-spacecraft combination for launch. In doing so the launch authority relies on a thorough technical, legal, and safety review process that typically starts years or months prior to launch. To obtain final clearance the spacecraft owner/operator and the launch authority will moreover need to obtain certain licenses and approvals by national authorities, irrespective of the size of the satellite, including CubeSats.

Article VI of the *Outer Space Treaty* establishes the provision that states are internationally responsible for national activities in outer space, whether carried on by governmental agencies or by nongovernmental entities. Moreover, state party to the treaty has to ensure adequate and continuous supervision of the activities. Thus, states are under the obligation to authorize private space activities, including CubeSat missions [11]. In the implementation of these obligations, states are free to define whatever practical measures they consider adequate and sufficient to ensure compliance. They can do so by defining licensing processes to be followed by the entity undertaking the space activity or by establishing a national legal framework that regulates the activities, mandating a competent national authority, such as a space agency or ministry, to authorize and supervise those private space activities.

Prior to launching the satellite, the developer must apply for the necessary licenses to obtain authorization from the state to undertake the space activity. Similarly to how driving a lorry loaded with dangerous substances requires a driving license and certain specific permits, operating a satellite requires a license to launch, communication licenses for the usage of uplink and downlink frequencies (see [Section 4](#)), and any additional licenses required for the specific mission, such as a remote sensing license and reentry license. As an integral element of the mission, the ground station and the spacecraft operators need to be licensed to execute the communication activities. The responsible government authority, before granting a license, will verify that the activity proposed by the licensee does not interfere with or by other means causes harm to other ongoing or planned activities and is carried out in compliance to applicable legislation.

Different licenses may require addressing different government authorities, and in some states the authority to be addressed may furthermore depend on the nature of the mission (e.g., low Earth orbit missions vs. geostationary or planetary missions, and commercial vs. academic missions). The processes and conditions for authorization thus differ per country and may depend on the envisaged mission, but they typically cover some or all of the following aspects:

- frequency coordination, out of band emissions and transmitting power;
- technical maturity of the system;
- safety factors;
- orbital lifetime assessment;
- space debris mitigation;
- collision avoidance capability;
- postmission disposal;
- reentry casualty risk;
- dual-use technology and export control;
- recourse for damage, insurance, waivers, and third-party liability;
- financial capacity;
- environmental factors.

Licensing: Practical recommendations for CubeSat developers

- Identify from the national space law, regulatory acts or practices the licenses to be obtained and additional requirements imposed on the activity (e.g., insurance and maneuvering capability).
- Identify which authorities manage the licensing requests and the requirements to be met by the licensee and the applicable processes to obtain them.
- Consider any relevant requirements in the design and testing of the spacecraft and ground station. For example, the requirement to be able to terminate all transmissions to ground (in case of interference) could result in the implementation of a *silent mode* in the operational architecture.
- Provide the inputs to the relevant authorities following the applicable processes. Once authorized the CubeSat is “ready to go” from an administrative/regulatory point of view. However, this does not represent the last contact with the administrative authorities, and all through the mission’s execution the project team and the administration will have to keep close ties. Ensure that the communication link works well and the project is under the correct regulatory scrutiny.

- Governments or organizations enacting new regulatory frameworks for national space activities are recommended to consult with the United Nations General Assembly Resolution 68/74 of December 11, 2013 on “*Recommendations on national legislation relevant to the peaceful exploration and use of outer space.*”

4 Radio-frequency registration

Radio signals from satellites naturally propagate from space into many states, necessitating worldwide frequency coordination among satellite operators. Defining the satellite and ground station operating frequencies during the preliminary design is crucial for the success of a mission, not only because it drives the spacecraft design parameters but also because the frequency spectrum and orbital positions are in fact limited natural resources that must be used rationally, efficiently, and economically [11].

The allocation of the radio-frequency spectrum is managed by the International Telecommunication Union (ITU), a dedicated agency of the United Nations that is responsible for global information and communication issues. Its *Constitution and Convention* [12] and the *ITU Radio Regulations* [13] constitute internationally legally binding rules, according to which also national regulations have to be developed. Subsequently and subject to the law applicable per country, national authorities may need to grant the final spectrum license. For example, in the United States, the Federal Communications Commission manages licensing processes on a first-come-first-served basis or via auction, after the ITU process has been successfully completed [14].

Per the *ITU Constitution*, member states can assign particular frequencies and orbital positions to their respective satellites, but they are obliged to avoid causing harmful interference to the frequency assignments that have been registered earlier, where noncompliance could result in a legal obligation to prematurely end the mission. On the other hand, a state that registers its satellite system using a particular orbital position and certain radio frequencies benefits from international recognition and is protected against harmful interference from latecomers to the registration process (“first-come, first-served rule”). It is in light of this policy that any space station (satellite) shall be fitted with devices to ensure immediate cessation of their radio emissions by telecommand, whenever such cessation is required under the provisions of the Regulations [15].

For CubeSats that are planned to be released to orbit in conjunction with other satellites, it is moreover valuable to set up a communication channel with other operators and to coordinate on the downlink and uplink frequencies. Sharing of information on contacts made with the spacecraft assists in the process to discriminate the different satellites from each other during early operations (see also Section 8.2), and for satellites with the same operating frequencies, it allows operators to request, for instance, to switch off the transmission channel of a copassenger during certain ground station passes to resolve any interference issues that may arise while the satellites fly in close proximity.

4.1 *Radio amateur frequencies*

Contrary to the understanding that many CubeSat developers have shown, satellite systems operating in the amateur-satellite service may be exempt from the full administrative fee that is charged for other filings, but they are not excluded from the frequency assignment notification and recording procedures [16]. Furthermore, when planning to operate in the amateur-satellite service, developers should coordinate additionally with the International Amateur Radio Union (IARU), an international confederation of national amateur radio organizations.

While the usage of radio amateur frequencies has been common practice for CubeSats of a different nature, it should be emphasized that the amateur service is reserved for applications of training, intercommunication, and technical investigations carried out by amateurs, that is, by duly authorized persons interested in radio technique solely with a personal aim and without pecuniary interest [13]. Moreover, transmissions between amateur stations of different countries cannot be encoded for the purpose of obscuring their meaning, except for control signals from ground to a satellite operating in the amateur-satellite service. Instead, with the exception of telecommands for critical satellite functions, all transmissions should be open for use by amateur radio operators worldwide. Practically, that implies that not only CubeSats launched for commercial services, but also CubeSat technology demonstration missions and scientific missions that do not comply with the earlier description, cannot qualify to operate in the amateur-satellite band.

Developers should thus consult the Radio Regulations to identify the frequencies that can both legally and technically be assigned to the mission. As a good practice, consultation with the IARU satellite advisor before initiating further processes allows the mission developer to assess whether or not the proposed frequency will interfere with other planned amateur-satellite services.

4.2 *Frequency allocation*

Frequency allocation and the licensing process are intended for the purpose of avoiding harmful interference between different applications when operative in the same geographic area. The ITU Radio Regulations provide a set of tables that describe the division of the global radio spectrum into bands allocated to particular applications, such as maritime radio navigation, meteorological services, space research services, radio astronomy, intersatellite links, and radio amateur services. For the allocation of frequencies, the world is furthermore divided into three regions, as shown on the map in Fig. 1, where each region has its own set of frequency allocations [13].

Furthermore, two categories of services are set according to the importance and the need for protection of the frequency allocation: primary and secondary. CubeSat missions, in most cases, fall into the category of secondary service. This implies that they can claim protection from harmful interference from other stations in the secondary service, but they cannot cause harmful interference to and cannot claim protection from harmful interference from stations of a primary service, even if assigned at a later date [13].

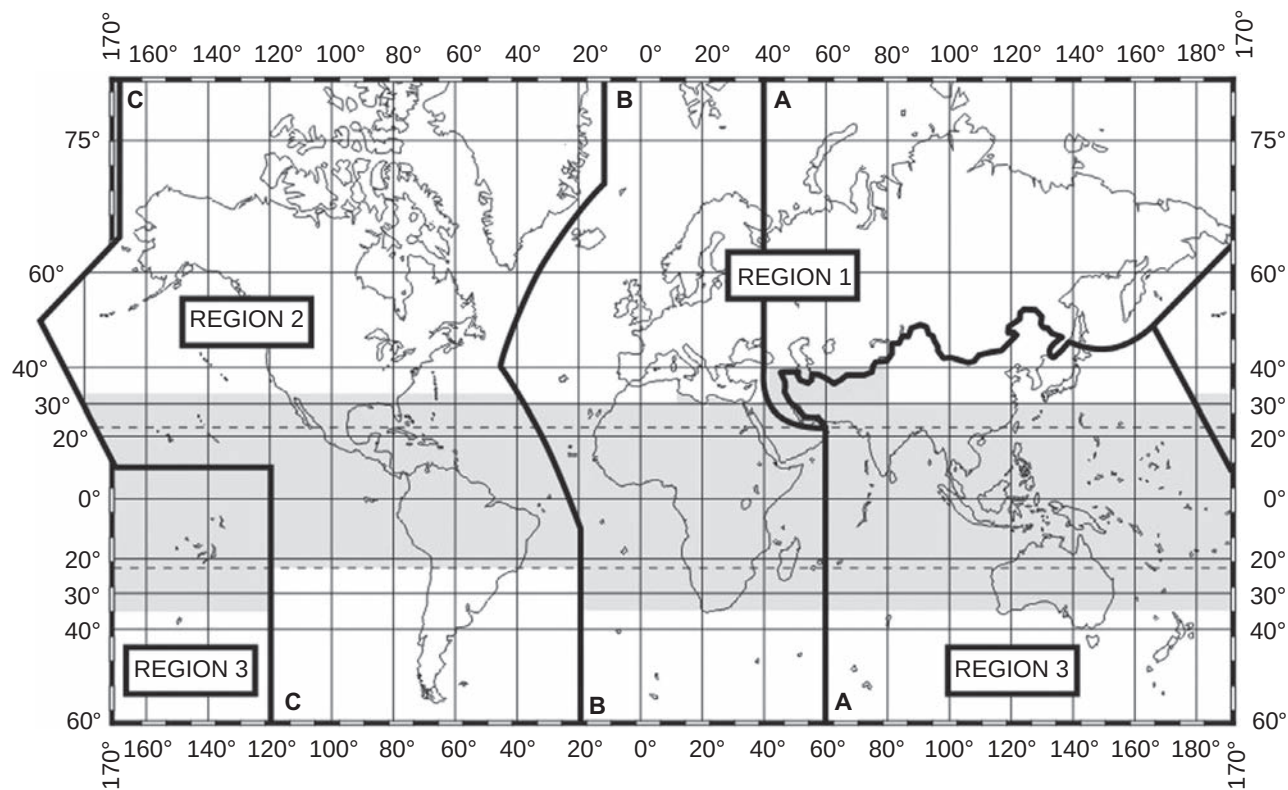


Fig. 1 ITU regions for the management of the global radio spectrum [17].

4.3 Frequency assignment notification and recording procedure

The process to obtain the recording of a frequency assignment in the Master International Frequency Register (MIFR) is described in Article 9 and Article 11 of the ITU Radio Regulations. [Fig. 2](#) outlines the frequency registration timeline, which essentially breaks down into the following key steps [18]:

- coordination with national administration responsible for radio communications,
- coordination with International Radio Amateur Union (in case of radio amateur–satellite service),
- publication in the Advance Publication Information (API) (ITU Radio Regulations Article 9 Section I),
- formal coordination with ITU (ITU Radio Regulations Article 9 Section II),
- notification (ITU Radio Regulations Article 11).

Many CubeSat missions are however exempted from the coordination process (Article 9.2), which only applies if the requirement to coordinate is included in a footnote to the Tables 9.11A-1 and 9.11A-2 of the ITU Rules of Procedure. On the other hand, Articles 9 and 11 are applicable also to the amateur-satellite services, as indicated in [Section 4.1](#).

A. Submission of Advance Publication Information (API/A)

Any administration bringing a satellite into use shall provide to the ITU Radio-communication Bureau a general description of the network for advance publication. The characteristics to be provided for this purpose are listed in Appendix 4 of the ITU RR, titled “Consolidated list and tables of characteristics for use in the application of the registration procedures” [13]. Examples of information to be included are the antenna beam characteristics, frequency bandwidth, emission power levels, out-of-band emissions, and ground station descriptions.

The API should preferably be published no later than 2 years before the planned date of bringing the CubeSat into use of, but given the shorter development cycles of CubeSats compared to larger satellites, providing the API at least 1 year before launch can be considered acceptable. It is however recommended to prepare the API as early as possible during the development phase to initiate the process, even if certain details normally included may not yet be known, such as the planned launch vehicle, launch date, and orbital parameters. In such cases ITU generally accepts the API without having these data frozen, and those shall then be communicated no later than upon submittal of the notification request (Step C).

Within 3 months after receipt of the complete information from the national administration, the ITU Bureau shall publish an API/A special section in its biweekly BR International Frequency Information Circular (IFIC).

B. Comments from administrations and API/B publication

After publication of the API/A, any administration that believes that unacceptable interference may be caused to its existing or planned satellite networks already registered can provide comments within a period of 4 months, which are published by

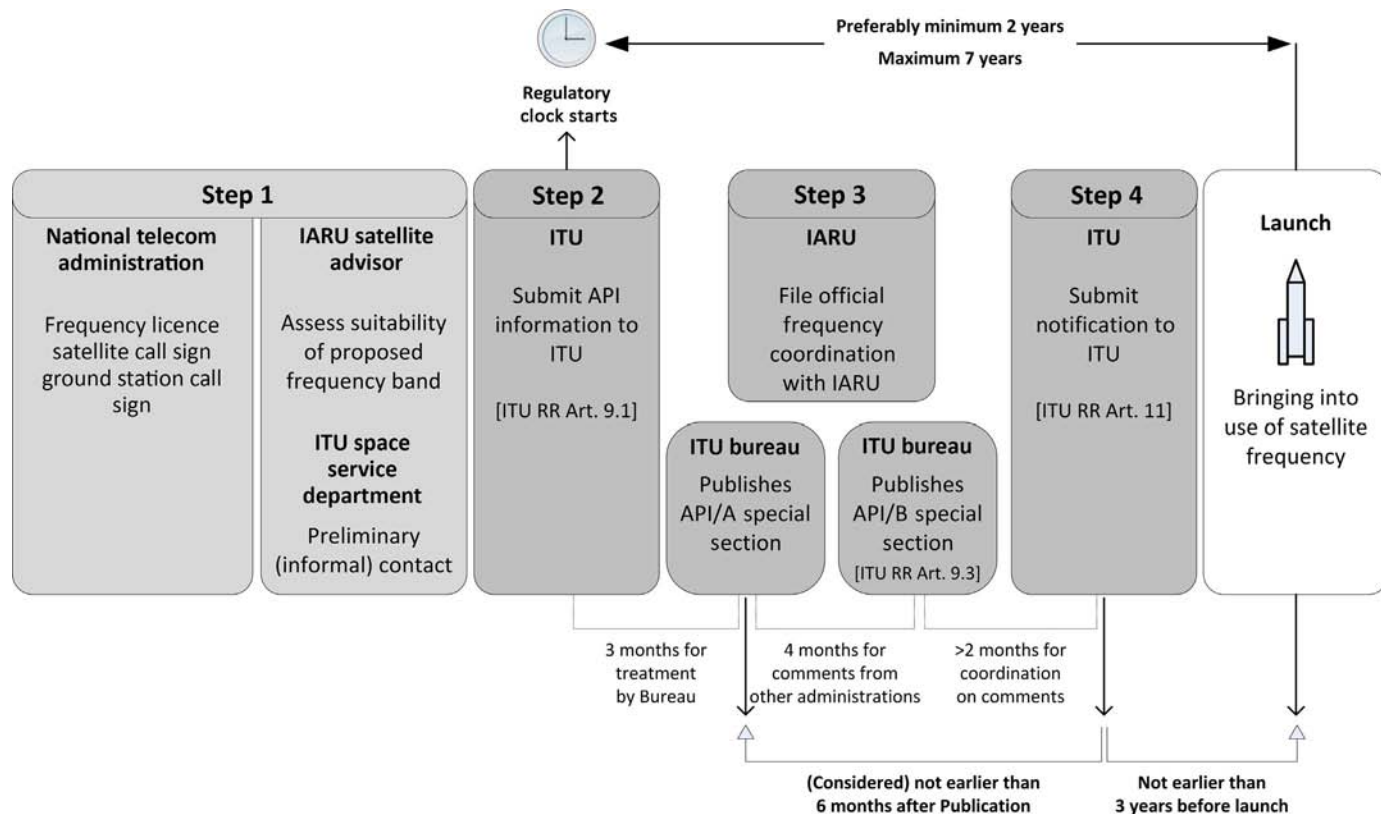


Fig. 2 Frequency registration timeline.

Source: Author's own figure.

ITU Bureau through an API/B special section. Thereafter, both administrations shall endeavor to cooperate in joint efforts to resolve any difficulties, with the assistance of the Bureau.

C. Submission of Notification

After the API process is completed and before launch or operations, the frequency assignment of transmitting and receiving satellites and ground stations shall be finally notified to the ITU Bureau, in accordance with Article 11 of the Radio Regulations. This should be done:

- not earlier than 2 months after the date of publication of the API/B (if any),
- not earlier than 6 months after the date of publication of API/A,
- not earlier than 3 years before the first use of the frequencies [13].

The notification information may however also be communicated to the Bureau together with the Advance Publication Information. In that case, ITU will consider the notification as having been received no earlier than 6 months after the date of publication of API/A. Finally, any frequency assignment to a space station provisionally recorded shall be brought into use no later than 7 years from the date of receipt by the ITU Bureau of the API information [13].

Communication with the ITU Bureau shall be performed by the responsible national administration only. If more than one state is involved, one state licensee must be agreed upon between them so that only one administration communicates with the Bureau.

Recommendations for CubeSat developers w.r.t. frequency licensing

In the case of radio amateur frequency use, developers must:

- o Informally consult the satellite advisor of the International Amateur Radio Union (IARU) to assess whether the proposed frequency is not interfering with other planned amateur-satellite services.
- o Request an amateur call sign for the spacecraft.
- o Ensure that the ground station and operators are licensed and have obtained a call sign.

The coordination request to IARU may be undertaken directly between the licensee (trustee of the ground station) and the IARU satellite advisor.

The latest template of the Frequency Coordination Request can be obtained from the IARU satellite advisor. The template advised the developers to:

- Prepare the Advance Publication Information (API) filing to the ITU Bureau, using the latest version of the ITU Bureau software for the capture and validation of the space notices available on the ITU website.
- Submit, via the national administration, the required data to the ITU Bureau.
- In case of radio amateur frequency use, at the time of submission of the API information to the Bureau, submit the formal Frequency Coordination Request to the IARU satellite advisor.
- Upon completion of the API process, follow the official filing procedure with the national administration to submit the Notification to the ITU Bureau (or communicate it to the Bureau together with the submission of the API).

Satellite developers can consult information considering actual/regulatory status of the API and Notification of their particular or any other satellite networks via the ITU-R Space Network List (SNL).

For an overview of the national Telecommunications Regulatory Bodies, refer to the ITU Global Directory.

5 Space object registration

Space object registration forms another important obligation, although here the prime obligation is on the part of the respective state, not on the CubeSat developer. The *Convention on Registration of Objects Launched into Outer Space*, briefly the *Registration Convention*, was adopted by the UN General Assembly in 1975 and entered into force 1 year later. It provides for a legal framework to maintain databases of all man-made objects launched into Earth orbit or beyond.

Firstly, states undertaking space activities shall maintain a registry for all their objects launched into space [10]. Evidently, this requires first and foremost to set up an adequate space object registry at national level—a requirement not yet fulfilled in all states. In particular, when a CubeSat happens to be the very first space object of a state, competences, administrative procedures, and the space object registry need to be established. This includes assigning a governmental body the responsibility over the national registry (usually the national space agency or space office or any other ministerial unit responsible for space activities) [11]. Secondly the secretary-general of the United Nations maintains a register that is openly accessible in which, for each space object, it records the following:

- the name of the launching state (or states),
- an appropriate designator of the space object or its registration number,
- the date and territory or location of launch,
- the basic orbital parameters (nodal period, inclination, apogee, and perigee),
- the function of the spacecraft,
- additional information as provided by the state of registry,
- date of decay/reentry (if applicable, through additional notification).

It is the responsibility of the launching state administration to furnish those data to the UN secretary-general as soon as practicable, using the model registration form that consists of four parts covering (a) information provided in conformity with the *Registration Convention*, (b) additional information relating to a change of operational status, (c) transfer of ownership, and (d) additional voluntary information [10,19]. The notification to the UN is to be sent by the competent authority of the state of registry through a diplomatic mission accredited to the United Nations.

The term launching state, as defined in the *Registration Convention* Art. I, refers to any state that launches or procures the launching of a space object, or from whose territory or facility a space object is launched. By definition, it is likely for CubeSat missions, as for a vast amount of missions, to involve multiple launching states. According to the *Registration Convention* Article II, in such case, it is up to the state parties to jointly determine who should register the object in its national registry and

furnish the data to the UN secretary-general. The state from whose territory a satellite is launched, for example, qualifies as a launching state, but it will typically not register the satellite if it has no obvious relation with the spacecraft other than hosting the launch base. More commonly the state of origin, on whose territory the activity is being undertaken, such as the state where the satellite operator is based, would be the one registering the object. The registration of a space object does not directly trigger liability for damage caused by that object (see [Section 9](#)), but indirectly, there is a clear link: in determining liability, which lies with the launching states, the state of registry will always be considered as only a launching state can register a space object. Certain launch authorities, for instance, those based in Japan, do request evidence prior to launch that the state of origin of the spacecraft is prepared to register the space object after launch. This practice is also being adopted by an increasing number of launch service providers on request of their governments.

International intergovernmental organizations conducting space activities and that have declared acceptance of the *Registration Convention*, such as the European Space Agency, may furthermore act similarly as a state within the context of the *Registration Convention* and may register objects on behalf of its member states [10].

Since the registries must contain details linked to the launch date and orbit parameters, the actual registration in the national registry and the notification to the UN secretary-general are foreseen to take place after launch, as soon as practically possible. It is nevertheless advisable for CubeSat developers and especially those for whom the CubeSat will be the first, or among the first space objects to be launched for that state, to contact the government authorities well before launch to coordinate on and prepare the actual registration. Space object registration not only is an administrative act but also has important legal implications. In particular, through the registration of a space object, a given state will carry jurisdiction and legal control over the satellite [11]. Obtaining a declaration prior to launch on the readiness of the state to register the satellite aids to mitigate the risk of dubiety over the legal status of the satellite after it has been placed in orbit.

Recommendations for CubeSat developers w.r.t. space object registration

It is recommended that CubeSat developers:

- Familiarize with the UN *Convention on Registration of Objects Launched into Outer Space* (accessible at <http://www.unoosa.org/oosa/en/ourwork/spacelaw/treaties/introregistration-convention.html>).
- Verify which information should be provided in the UNOOSA *Information Form* to be submitted to the UN secretary-general. A model registration forms has been made available in all official languages of the United Nations on the UNOOSA website.
- In case there are multiple launching states, clarify which launching state will register the satellite. Consult with the *United Nations General Assembly Resolution on the "Application of the concept of the 'launching State' (UNGA Res 59/115)* for the identification of all launching States for the mission."
- Identify which is the competent authority in the state, establish contact with the authority during the satellite design or development phase, and share relevant information about the satellite activity.
- Obtain confirmation on the state's readiness to register the space object upon launch in the national and UN registries.

- Upon launch, provide the required information such as exact launch date and orbital parameters, through the appropriate forms and channels, and follow-up with the national authority on the registration of the CubeSat in the national and United Nations registries.
- Upon reentry, whether following a deorbiting maneuver or natural decay, inform the authority to notify the UN secretary-general, as to update the object's status in the registries.

It is noted that states not party to the Registration Convention can still voluntarily provide registration information on their space objects [20].

The *UNOOSA Handout on Small Satellites*, freely accessible online, serves as a good reference for further guidance on the space object registration.

6 Safety and cleanliness

The authorization to launch from the state and the acceptance for launch by the launch authority or launch service provider will rely, among other conditions, on a positive outcome of various safety reviews. Not only should the personnel's safety be guaranteed during the spacecraft processing and preparation for launch, but also the launch authority is tasked to reassure that the CubeSat cannot pose a threat to the good integrity of the launch site, of the launcher, and of any other spacecraft onboard the launcher, for instance, through radio-frequency interference, inadvertent activation of the spacecraft, breakup resulting in release of particles or fragmentation, and materials outgassing.

Specific aspects of safety come into play throughout the various program phases, but most of them should be considered starting from the detailed design stage, where they may already have an impact on engineering trade-offs and design choices. Use of pyrotechnics, pressurized vessels, and propulsion evidently leads to more stringent safety procedures and may pose important constraints on the spacecraft handling on ground and in space. But the selection of materials, the type of batteries, and the architecture of the power management system needs to be defined with attention for the environment in which the satellite will need to operate on ground, during launch, and in orbit.

This includes the limitations posed by environmental legislation that limits the use of potentially harmful materials. For example, antenna systems for numerous CubeSats have been made using beryllium-copper alloys, due to their good combination of mechanical and physical properties and formability. The hazardous nature of beryllium, however, requires it to be processed with adequate care for the health risks that it poses if during machining activities particles are inhaled or swallowed. Similarly, Alodine 1200, widely used as conversion coating on aluminum substrates in spacecraft and terrestrial applications, is subject to restrictions that have recently been imposed by certain safety standards due to the associated health risks. It is thus important for CubeSat developers to select materials with awareness on the safety risks they may imply or the constraints they pose with regard to their manipulation and handling. Directives and standards can be consulted to assess the risks in the selection of materials, such as the European registration, evaluation, authorization, and restriction of chemicals (REACH) regulation and restriction of hazardous substances (RoHS) directive.

Other elements that are subject to in-depth safety assessments, in particular by the launch authority, include the configuration of the electrical inhibits to keep the satellite unpowered during launch; the CubeSat's onboard batteries; and any deployables such as antennas, solar panels, or booms must not be deployed prior to deployment of the satellite. A CubeSat is generally not allowed to transmit during launch or soon after its deployment to orbit to minimize the risk of electromagnetic interference to the launcher or other spacecraft. Moreover, the CubeSat should not deploy any appendages until after its release to orbit to avoid mechanical failures.

Battery systems, as they need to withstand extreme temperatures and extremely low pressures, should be equipped with sufficient protection mechanisms and require extensive testing in environmentally representative conditions. Requirements for battery testing differ depending on the type of launch, but in particular for batteries onboard CubeSats that are planned to be launched from or transported to manned space systems, such as the International Space Station, the qualification process is more stringent and may include destructive testing. In relation to the applicable rules on space debris mitigation, certain organizations furthermore require assessments related to the risk for in-orbit explosions and the risk for fragmentation of the satellite or its parts in orbit, and for such assessments the batteries and its protection mechanisms (e.g., overvoltage protections and pressure vents) are typically investigated. It is recommended that the developers select and procure batteries with flight heritage or that have been qualified for the use case by the launch service provider, or otherwise to be aware of the stringent safety and qualification requirements that are typically imposed on batteries.

In selecting materials and components, it is also imperative to consider the extreme temperatures, vibration, and pressures during launch and in orbit. An important aspect to consider is the outgassing of materials, which for any spacecraft must remain below acceptable levels specified by different engineering standards and interface documents. Outgassing is the release of gaseous material from a specimen under high vacuum conditions. It may pose a contamination threat to the satellite itself or to other passengers onboard a shared launch, due to the risk that outgassed material deposits on sensitive surfaces such as optics, with potentially high impact to the mission. For this reason, launch authorities may require evidence of bakeout, a process in which the hardware is placed in a high-temperature environment, usually in vacuum, thus accelerating its outgassing rates to reduce the content of molecular contaminants present in the material. Several major space agencies provide publicly accessible databases of materials, components, and parts that have been tested and for which the outgassing properties are known, which CubeSat developers can consult during the selection process.

Furthermore, due to the rapid pressure drops that the CubeSat faces during liftoff, it is important to enable sufficient venting on the satellite and its deployer to avoid that air traps could result in bursts or explosions with a potentially detrimental impact to the launcher. The minimum venting area depends on the compartment volume, and engineering standards describe requirements on venting that can be readily implemented.

Finally the overall cleanliness of the satellite during ground processing must be guaranteed as good practice to prevent issues at spacecraft or lower level and for similar reasons as above vis-à-vis copassengers onboard the same launch. During storage and assembly, integration, and verification (AIV) activities, the CubeSat should be subjected only to a controlled, clean environment with controlled temperature and

relative humidity. Cleanroom class ISO-8 (class 100,000) or better are generally considered adequate for CubeSats, if no more stringent requirements are imposed by the payload or external factors.

Recommendations for CubeSat developers: Safety and cleanliness

- Identify safety requirements applicable to the mission, either imposed by law, by the launch authority, or other, and consider the requirements as of the detailed design stage.
- Obtain access to a cleanroom class ISO-8 or better to process all flight hardware during the assembly, integration, and verification (AIV) phase.
- During manipulation of electronics, ensure operators and hardware are connected to the same electrical ground, among others, by use of antistatic mats and antistatic wrist bands.
- For the transport of flight hardware, make use of antistatic wrapping or bags and place them with the hardware inside a suitable transport container or flight case that limits temperature and humidity fluctuations, and exposure to shocks.

7 Export control

States implement export control regimes as part of their foreign policy and national security strategies to control, restrict, or embargo the cross-border transfer of potentially sensitive technology. Export control aims to limit the proliferation of ballistic missiles, weapons of mass destruction, and dual-use technology [21]. Furthermore, economic motivations may play a role in the regulations set forth by governments. Due to its civilian and potential military applications, spacecraft technology and hardware has an inherent dual-use nature. Communications, navigation, and remote-sensing capacities enabling peaceful satellite missions are intrinsically not different from those required for intelligence services and military end use. Their export is bound by restrictions that should be appreciated during the satellite development process, regardless of its size and function and thus also when it concerns a nonmilitary CubeSat mission.

Controls on exports cover the transfer from one entity to another of domestic hardware, software, services, and intellectual property that involves the design, development, use, or operation of the technology in question. Thus the exchange in whatever form and by whatever medium of technical information such as design data, technical drawings, user manuals, or even processes and skills can be subject to export control regulations, the violation of which can lead to significant criminal and civil penalties to companies or individuals [21]. As an example, prior to releasing restricted technical data on launcher interfaces to a foreign customer, US launch service providers and their customers must obtain a technical assistance agreement from the US State Department, and similarly, in the European Union, the transmission of technical data to non-EU launch service providers, for the integration of the spacecraft onto the launcher, requires prior authorization by the competent authorities.

At an international level, nonbinding multilateral instruments are in place, including the Missile Technology Control Regime (1987), the Wassenaar Arrangement (1994), and the Hague Code of Conduct against Ballistic Missile Proliferation (2002). In addition to the multilateral treaties, various states define the implementation of export control regimes through binding domestic legislation, typically based on a periodically updated list of controlled items [22].

Even if most satellite technology is subject to export regulations, not all dual-use items require explicit authorization. Due to the relatively complex nature and the multiplicity of regulations among different states, CubeSat developers are encouraged to seek counsel in the domain as to mitigate cost impacts, schedule overruns or sanctions. A concise description of the practices in the European Union and the United States is provided here as example.

The EU export control regime provides for relatively swift authorization procedures, aiming to stimulate a space industry with a rather high share of export sales [21]. While the final authority is with the EU member states, which may complement or expand them, a set of common rules governs the control of military technology and equipment through the Common Military List. Due to their technical specificities, satellites exported outside the EU for launch from a non-EU territory, regardless of size and function and thus including CubeSats, require prior authorization by the competent member state authorities, which is also needed for the exchange of related technical data. It may moreover be expected that suppliers of parts and subsystems may request a formal declaration on the nonmilitary end use of the ordered items from their customers, as such end use would otherwise impose the supplier to apply for a specific export authorization.

In the United States, various regulations cover items defined by specific categories, i.e., munitions, and the competence to issue export licenses is diffused among different government agencies. Most noteworthy are (a) the International Traffic in Arms Regulation (ITAR) including the US Munitions List, which covers all military spacecraft and of which items cannot be exported to countries under US embargo, and (b) the less-restrictive Export Administration Regulations (EAR) including the Commerce Control List, which nowadays covers most commercial space technology, easing the regulatory burden on many CubeSat operators.

The export of a nonmilitary CubeSat to a non-US territory will usually require the submission of a license application within the scope of ITAR, while at lower level the export of the vast majority of off-the-shelf components (COTS) are subject to the EAR but do not require an actual license, easing their export. EAR licenses cover both hardware and technology; export of ITAR-controlled technology is more burdensome, where hardware export necessitates a license, while the release of technical data or services are regulated via the earlier mentioned Technical Assistance Agreements and Manufacturing License Agreements. Integration of a US satellite under EAR control into a non-US launch vehicle and any related exchange of technical data is always subject to ITAR [21].

8 Space debris mitigation, collision avoidance, and reentry casualty risk

As introduced by Abercrombie and Ostrom in Chapter 20, the paradigm shift toward smaller satellites, the vast increase in the number of satellites launched, together with the numerous large-scale constellations under development have given rise to increased concerns by experts on the threats posed by space debris. In-orbit collisions

or breakups and the eventual manifestation of the Kessler syndrome through a cascade of debris collisions pose significant threats to on-going and planned space activities that call for responsible behavior by all space actors, including CubeSat operators. As indicated in [Section 3](#), compliance to space debris mitigation requirements and safe reentry are part of the safety approvals required to obtain authorization for the mission.

8.1 *Space debris mitigation*

Space debris mitigation comprises the practices and procedures aiming to

- limit the release into orbit of launcher upper stages and satellite parts during nominal operations,
- avoid intentional destruction of satellites,
- limit the probability of accidental collisions,
- minimize the potential for breakups during operations or after the end of mission,
- limit the long-term presence of satellites and launcher stages in certain protected orbital regions.

By introducing certain basic concepts in the CubeSat design, developers can help in preventing the generation of space debris. As a first step the release of objects or particles larger than 1 mm should be avoided. For the nominal operations, this is achieved by design, by ensuring that no parts such as deployable retainers, lens caps, and separation bolts are intentionally released. This also applies to the burn wire often applied onboard CubeSats as thermal knife in a deployment mechanism.

Release of parts or entire fragmentation of the spacecraft as a consequence of anomalies must also be prevented. This is achieved among other methods through good structural design, adequate selection of fasteners, the use of secondary fastening techniques, and by performing an assessment of the in-orbit breakup risk. Also important to this purpose is the design of the electrical architecture, including the selection of battery boards with adequate protection mechanisms to prevent pressure buildup, battery overcharging, thermal runaway, and short circuits.

Prior to launch the undertaking of a suitable verification program (e.g., testing the battery protection mechanisms in a representative environment) should subsequently provide confidence of the design and the tolerance for failures. Finally, successful disposal at the end of life comprises the deorbiting maneuver or natural orbital decay and spacecraft passivation. Passivation is achieved through the depletion of all energy stored on-board to avoid the satellite undergoing breakup after the end of mission. Propellant tanks can be depressurized by means of venting and depletion burns. Batteries can be discharged and disconnected to prevent explosions as they degrade. Should passivation of the power system not be performed, or safety inhibits fail and pressure buildup does lead to explosion of batteries, physical encapsulation of the batteries inside the spacecraft may further help to prevent in-orbit fragmentation.

In the selection of the launch and orbit, it should be confirmed by analysis that the orbital lifetime will remain below prescribed thresholds and with minimal collision risk. Various standards demand the removal of the spacecraft no more than 25 years beyond the end of the operational mission for satellites in the low Earth orbit protected

region (up to 2000 km). In practice, however, satellites with lower reliability are recommended to start the clock at the launch date. Furthermore, considering also the reliability of utilized parts and depending on the type of application, responsible operators may target far lower orbital lifetimes that approach more closely the operational life span of the satellite. The effective orbital lifetime will depend, among other variables, strongly on the solar activity (that affects atmospheric density and in turn influences orbital drag effects) and the area-to-mass ratio of the spacecraft (and resulting drag). To a much lower effect, the satellite attitude can also play a role as the orientation with respect to the velocity vector can affect the cross-sectional area. As a rule of thumb, disregarding any station-keeping or deorbiting capabilities and assuming circular orbits, one-, three- and six-unit CubeSats can be released at altitudes, respectively, below ca 550 km, 575 km, and 625 km, to comply with the 25-year ruling [23].

If the movement of a space vehicle to nonprotected regions or their reentry does not happen naturally in the space environment, active deorbiting capabilities such as drag augmenting devices or propulsion can be used. The technology associated with such capabilities, however, needs to be highly reliable in order to demonstrate that the mission can comply with the 25-year rule. The effective functioning of an experimental drag sail, for example, should not be vital to the successful timely disposal from the protected region, as the reliability of such experimental sail would be difficult to demonstrate prior to its in-orbit validation.

As the development of large-scale constellations gives rise to increased concern over the risks for in-orbit collisions, commercial operators may play an important role in implementing new practices that could evolve into technical standards over time. This can be expected in the areas of in-orbit serviceability, space traffic management, and possibly other areas. New custom practices will arise beyond the currently adopted guidelines and standards, which may help to mitigate the further growth of debris. Those could include the installation of standardized external fixtures to interface with active debris removal satellites [24], the use of retroreflectors on external surfaces to facilitate tracking, new deorbit and reorbit techniques, and automated collision avoidance maneuvering. In the future a so-called space sustainability rating may also offer an innovative means to encourage the responsible behavior of organizations with respect to debris [25].

While space debris mitigation is not mentioned in the five space treaties introduced in Section 2 and while in some cases the standards may not be legally binding, the sustainability of operations in the protected regions calls for operators to act responsibly and with consideration for the fact that also space is a shared resource in which one's behavior may directly affect current and other future operators.

8.2 Collision avoidance

As part of the mission analysis, collision avoidance should be addressed from various perspectives. First, in selecting the target orbit during the initial mission analysis, developers can take into consideration the probability of impacts with other objects in a given orbit throughout the spacecraft's orbital lifetime. Second and in particular for shared launches with multiple satellites, as usually the case for CubeSats, the

launch authority will perform its own mission analysis to define a deployment scenario that minimizes the risks for cross collisions between different satellites and the launcher during the separation. Third, during the entire orbital lifetime of the satellite, close approach events with other space objects, including other active satellites or inactive objects or other forms of debris, should be handled with maximum care. Collision avoidance processes are generally performed as best practice and are not required by most standards. Recent NASA and ISO standards on space debris mitigation, however, do demand that satellites with maneuvering capability must be operated, for instance, by making use of its limited propellant to maneuver in order to actively limit collision risks during the entire lifetime [26,27].

The world standard provider of space surveillance data is the US Combined Space Operations Center (CSpOC). This center conducts conjunction assessments for all active spacecraft against all catalogued satellite objects within defined threat thresholds [28]. At the same time, CSpOC maintains an operators' contact list and notifies operators of close approach events considered to pose a collision risk.

Prior to launch, operators are strongly encouraged to register with CSpOC. This enables the operators to be contacted in case of conjunction identification, that is, when the satellite is predicted to closely approach another space object. To define the course of action, operators of any involved active satellite subsequently can assess the associated risk—or rely on a commercial or governmental service provider to do so—typically based on the miss distance. When achievable, such action typically consists of a collision avoidance maneuver, using propulsion, or in case of CubeSats sometimes using attitude control systems to adjust the ballistic coefficient and resulting atmospheric drag, to cause a small change to the orbit altitude. Uncertainties related to the miss distance and direction together, however, make the latter less trustworthy. Moreover, the less precise orbital fidelity and often more limited pointing precision makes avoidance maneuvers more challenging [5]. Evidently, in case of two active satellites, maneuvering of either one or the other spacecraft requires direct coordination between both operators. Per the 2019 ISO standard on space debris mitigation [29], having a collision avoidance capability becomes mandatory in certain orbital regimes. However, CubeSats often do not have such capability and in most cases may not implement it in the near future. It is important to note that also if the satellite has no collision avoidance capability, it is valuable to subscribe to the CSpOC service and to share maximum data in case of a notification, if only about the noncapability to maneuver, as any information may prove crucial in the days and hours before a close approach event.

Collision avoidance efforts require first and foremost up-to-date information related to the ownership of active space objects by operators. In particular, when launching and releasing multiple payloads, as often the case for CubeSats, it has proven very challenging to discriminate the satellites from the ground until operators manage to establish contact and notify CSpOC and each other on the successful identification of their satellite. This in turn may help other operators eliminate candidates and point their ground antennas in the right direction on subsequent passes of the batch of objects. There are several cases of cluster launches where it took weeks or even months for all objects to be uniquely identified. Also here, precoordination and

sharing of data during the launch and early operations phase facilitate fewer complications. Additionally, new initiatives are being tested such as the use of either optical (laser) or radio-based “license plates,” being small low-power devices attached to the satellite that regularly transmit an identification beacon, while relying only on their own power generation and transmission system independently from the platform [30].

8.3 Reentry casualty risk

Due to atmospheric drag a satellite in low Earth orbit eventually reenters the Earth’s atmosphere at the end of its orbital life. The process subjects the satellite to extremely high temperatures—caused by friction — causing the structure to break up under the extreme loads due to the rapid deceleration. If the disintegration of the satellite is not complete and certain parts survive the extreme temperatures during reentry, they will impact the ground and potentially result in casualties or damage to property, for which the satellite operator can be held liable. Commonly adopted standards require that the risk for serious injury or death due to impact must be demonstrated below 1 in 10,000, which in practice and unless of complete disintegration requires controlled reentry through high-reliability deorbiting maneuvers, guaranteeing that scattered debris will only impact ground over unpopulated regions [31,32]. Software tools and models freely made available by different space agencies can be used to analyze the likelihood of survival of parts, and the risk of casualties in case parts impact the ground. Unless specific mission needs dictate otherwise, CubeSat developers are recommended to design for demise, that is, to design the satellite such that it will be destroyed completely during its atmospheric reentry, thus ensuring compliance without requiring further models and without causing additional operational constraints. Detailed analyses can be performed with different software packages made available by space agencies. For CubeSats up to three units, it is generally considered sufficient to perform a review of design to confirm that all materials will fully burn during reentry. The use of hazardous chemical substances or radioactive materials requires more in-depth safety analyses.

Recommendations for CubeSat developers w.r.t. collision avoidance and space debris mitigation and reentry

- Starting from the spacecraft preliminary design, refer to Space Debris Mitigation regulations and incorporate them into the design. As the development matures, provide evidence of compliance through reviews of the design, analyses, and tests. The main points of focus are the following:
 - (1) the overall design and operations that shall not foresee any release of objects;
 - (2) the on-orbit lifetime, which shall remain below 25 years and practically is targeted to remain as short as possible;
 - (3) the risk for in-orbit breakup, which should be demonstrated to be sufficiently low through, for example, battery cell protection features, acceptable operational battery temperatures, and encapsulation reducing the risk for fragmentation;
 - (4) the reentry casualty risk that shall be demonstrated to be lower than $10E-04$ through an assessment of used materials, the spacecraft configuration, and dimensional features.

- A collection of national space debris mitigation standards can be consulted on the website of the United Nations Office for Outer Space Affairs.
- Prior to selecting the orbit for the mission, make use of one of the freely accessible tools to perform an orbit propagation analysis and to verify how many impacts can be expected in a given orbit. Examples of tools are the NASA Debris Assessment Software (DAS) or the MASTER and DRAMA tools of the European Space Agency. Reentry should be ensured within maximum 25 years.

Repeat the analysis with multiple solar activity models and with both nominal and worst-case cross-sectional area values to decrease the influence of uncertainties.

- Use those same tools to verify the risk for parts of the satellite to survive reentry and the forthcoming casualty risk over certain areas (assumed 0 in case of complete disintegration). For one-, two- and three-unit CubeSats, review of the design can be sufficient and may not require further analysis.
- To be notified of any close approach event, prior to launch register with the Combined Space Operations Center (CSpOC). Any operator not under severe embargo by the United States can sign up for free.
- Prior to launch, define the risk threshold for the miss distance that will trigger a certain action, such as a maneuver (in case of maneuvering capability).
- Establish contacts with operators of other spacecraft on the same launch and maintain a communication channel during early operations. Particular attention should be given to copassengers that are operating in the same or very near frequency bands.
- Upon establishing contact with or identification of the satellite in orbit, notify CSpOC and possibly copassengers to support their efforts to identify other spacecraft in close proximity.

9 Third-party liability and insurance

Although a CubeSat is a relatively small and lightweight space object, it cannot be ruled out that it may cause damage, be it to other space objects or, as unlikely as this may be, to objects or people on the surface of Earth or to aircrafts in flight. International space law provides for a unique third-party liability regulation, in particular in relation to damages caused on the surface of the Earth: launching states are absolutely liable for such damages on Earth, or to aircraft in flight, that is, independent of a fault; for damages in outer space, launching states are only liable when at fault (e.g., in the case of wilful misconduct). Although this chapter does not discuss the liability regime of space law including its preconditions and implications in detail, it shall be pointed out that any CubeSat launch automatically triggers the international liability of at least one state (the state of registry) and possibly (and simultaneously) more. As mentioned in [Section 5](#), in the determination of liability, the state of registry will always be considered.

It is therefore vital to clarify the liability framework for a given mission between all involved actors at private, national, and, possibly, international level ahead of launch. Licensed operators must also be aware that, depending on the national regulatory framework under which they operate, they might be required by their government to cover part of a possible compensation of third parties injured or affected by the space object under their control. Likewise, national space laws may require the operator—not at least for that very purpose—to take out insurance and provide evidence of

coverage before granting the mission authorization. Insurance for a small satellite mission is not mandatory in every state, however. Some national frameworks—like the Belgian Space Act—do not provide for an insurance requirement but allow for a governmental decision to impose insurance coverage on a case by case basis. Vice versa, other national space laws—like the Austrian Space Act—go the opposite way, that is, having a general insurance obligation that can be waived if a space mission is in “public interest” (e.g., scientific and educational missions). Whether domestic law requires insurance or not, it is imperative for each small satellite project to be adequately informed about the legal requirements, implications, and, ultimately, consequences of sending a satellite into outer space [11].

Recommendations for CubeSat developers: Third-party liability and insurance

- Clarify the liability framework between all involved actors (private, national, and international) ahead of the launch service procurement.
- Coordinate on the obligation status for insurance with the relevant authorities, as a measure contributing to mitigate negative surprises “down the road.”
- Ensure national authorities are fully informed on the mission during the development phase, by contacting the national space agency, or the government administration responsible for space activities.

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